Topological Quantum Field Theory

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Abstract

In this paper, we analyse closed, oriented Topological Quantum Field Theories $Z :\to$, where we define to be the category of closed manifolds with a given orientation. This varies from the more common type of TQFTs (ones without any structure enforced on the manifolds besides, say, smoothness), but doing this makes dealing with TQFTs nicer because we show that we have a rigid, pivotal structure on , and we do not lose much generality. After proving some properties about this type of TQFTs, we end by proving that $(0+1)$-dimensional TQFTs are essentially the same as finite dimensional vector spaces and hence are boring, however we acknowledge how it gets difficult very quickly to study higher dimensional TQFTs.

1 Introduction

Topological Quantum Field Theory is an area of research that has recently become popular and useful amongst both mathematicians and physicists. The idea is we define a TQFT to be a functor $Z :\to$, which encodes the geometry of space and time into vector spaces of physical states and observables[2]. Here is the category of n-dimensional manifolds, with potentially some chosen structure such as orientation or compactness. If we truly want to do physics, then we can replace with Hilbert spaces over $\mathbf{C}$[3], and include a lot more geometry in $_n$. TQFTs have be shown to be useful in quantum computing and potentially for creating a theory of quantum gravity[2], and also can be used to give invariants of knots and manifolds.[1]

Definition 1.1 (TQFT) *A $(n+1)$-dimensional (oriented, closed) Topological Quantum Field Theory is a symmetric monoidal functor $Z :_n\to$.* [1]

There is a lot of implicit information given in this definition which we will start to unpack now. Firstly, we need to define the category $_n$. The objects in are defined to be oriented, closed (compact without boundary) n-dimensional manifolds (here and throughout the rest of the paper, manifold means smooth manifold). A morphism $M : E \to F$ is an equivalence class of oriented, compact cobordisms from E to F. A cobordism $M : E \to F$ is an $(n+1)$-dimensional manifold such that there exists injections $i_E : E \to \partial M$, $i_F : F \to \partial M$ such that the map

$$\overline{i_E} \sqcup i_F : \overline{E} \sqcup F \to \partial M$$

is an orientation preserving diffeomorphism. Here $\overline{E}$ denotes E with its orientation reversed and $\overline{i_E}$ denotes the associated orientation reversed map. Essentially, what we are describing is that M is an $(n+1)$-manifold with two boundary components, one diffeomorphic to $\overline{E}$ and one diffeomorphic to F. So we treat M as a map from E to F by thinking of M as describing a time evolution of E changing into F where the extra dimension of information described in M can be treated as the time coordinate. Thinking of cobordisms in this way, it is natural to define the composition of maps $M : E \to F$ and $N : F \to G$ by gluing M and N together along F. That is, $N \circ M = N \sqcup_F M$. Intuitively, we think about this as E evolving into F by M and then F evolving into G by N. However, there are some subtle details that need to be attended to in this gluing process, but first let us describe the equivalence relation mentioned above.

We define two cobordisms $M : E \to F$ and $M' : E \to F$ with associated inclusions i_E, i_F and i'_E, i'_F respectively to be equivalent if there exists an orientation preserving diffeomorphism $\psi : M \to M'$ such that the following diagram commutes.

$$\begin{tikzcd}
& M \arrow[dd, "\psi"] & \\
E \arrow[ur, hook, "i_E"] \arrow[dr, hook, "i'_E"'] & & F \arrow[ul, hook', "i_F"'] \arrow[dl, hook', "i'_F"] \\
& M' &
\end{tikzcd}$$

That is to say, two cobordisms $M, M' : E \to F$ are equivalent if and only if we have an orientation preserving diffeomorphism $\psi : M \to M'$ such that it does not matter whether we include into M, then apply ψ, or whether we just include into M'. Thinking of a cobordism as time evolution, then this equivalence relation is a very natural notion of having two time evolutions being essentially the same.

When we defined composition of two morphisms as gluing two cobordisms $M : E \to F, N : F \to G$ along their common boundary, there is a subtle detail hiding in the fact $N \sqcup_F M$ has a smooth structure compatible with the orientation of M and N. It is a non-trivial fact that the colimit $N \sqcup_F M$ exists in the category of topological spaces and can be used to construct a unique (up to non-unique diffeomorphism) smooth structure on $N \sqcup_F M$.[1] Since our cobordisms are defined only up to diffeomorphism, the fact that this smooth structure is only unique up to a non-unique diffeomorphism is a good enough classification for our purposes. However if we were to weaken the structure in our definition of , then we would have to be more refined in our definition of composition by gluing.

Now that we have defined , we notice that our definition of a TQFT asserted that $Z :_n \to$ is a symmetric monoidal functor. This implicitly states that $_n$ is a braided monoidal category with a symmetric braiding (we already know that is monoidal with the usual tensor product, and has a symmetric braiding). The natural choice for the tensor product in of $E, F \in$ is taking the disjoint union:

$$E \otimes F := E \sqcup F.$$

From previous knowledge of the disjoint union, this is strictly associative and so there is no need to check the pentagon diagram. Further, it is clear that the tensor unit in the category is the empty set $\emptyset$, and the obvious isomorphisms $\lambda_E : \emptyset \sqcup E \to E$, $\rho_E : E \sqcup \emptyset \to E$ make the triangle diagram commute. So is a monoidal category.

This category has a symmetric braiding because for any $E, F \in$, we have identity morphisms given by $M = E \times [0, 1]$, $M' = F \times [0, 1]$, which are just cylinders on E and F. If we take these cylinders M and M', put them next to each other and twist them around each other (by one half twist), then we have an isomorphism from $E \sqcup F$ to $F \sqcup E$. This construction is a symmetric braiding since the composition of two twisted cylinders is diffeomorphic to two untwisted cylinders.

Figure 1: Demonstrating the braiding in .

We have defined and shown it is a symmetric monoidal category. Notice that we are restricting ourselves to oriented and closed TQFTs, where instead we could have not chosen any structure for our manifolds in . The reason we have done this is because we have some nice properties about these types of TQFTs with very approachable methods to prove them. Additionally, we have not lost too much generality or practical application from this restriction.

2 Rigidity of

Proposition 2.1 *Let $Z : \to$ be a TQFT and $E \in$ an object of . Then $Z(E)$ is finite dimensional and $Z() \cong Z(E)^*$, where denotes E with the reversed orientation.*[1]

This proposition (which we will prove later) arises from giving our manifolds an orientation, which actually gives a rigid structure on . Since $E \in$ is oriented, we have a natural choice of a dual of E, namely $E^* = \overline{E}$. The evaluation map $_E : E\sqcup \to \emptyset$ comes from taking the disjoint union of E and and connecting them by a cylinder that runs above them. Here the boundary at the bottom of the cobordism is the source object in and the boundary at the top is the target object.

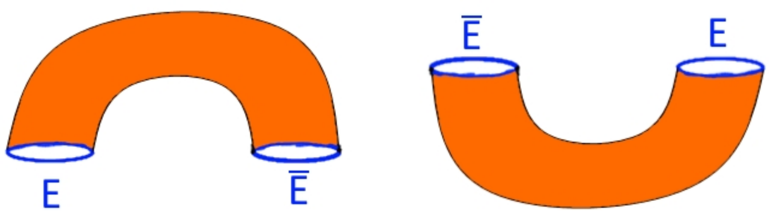

Figure 2: The evaluation and co-evaluation map in respectively.

In this case, the boundary at the bottom is $E\sqcup$ and the boundary at the top is $\emptyset$, so this is a map from $E\sqcup \to \emptyset$. Further the co-evaluation map $_E : \emptyset \to \sqcup E$ comes from taking the disjoint union of and E and connecting them by a cylinder that runs below them.

To see that these are evaluation and co-evaluation maps, we note that our cobordisms are invariant (as morphisms in) under certain orientation preserving diffeomorphisms and so we can literally take the composition $(_E \otimes \mathrm{id}_E) \circ (\mathrm{id}_E \otimes _E)$ and straighten it out to get the identity on E. This is similar to the string diagrams representing ev and coev, and is depicted below.

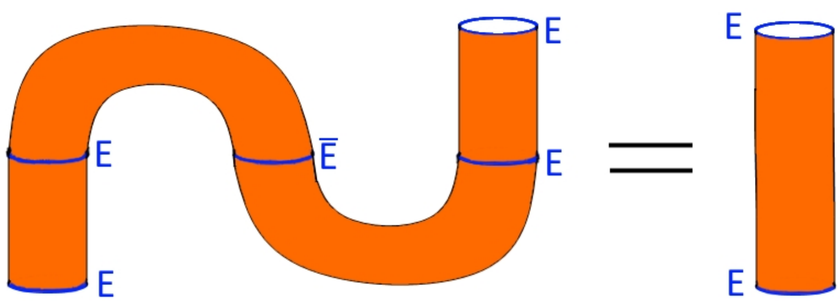

Figure 3: Showing that we can straighten out the ev, coev composition to the identity.

Similarly, we can do the same for $(\mathrm{id}_{\otimes E}) \circ (_E \otimes \mathrm{id}_)$ to get the identity on . Since we have a symmetric braiding between $E\sqcup$ and $\sqcup E$, we can see that has both left and right duals. Thus, is a rigid category. Further, since

E with its orientation reversed twice is just itself, we can give a pivotal structure by choosing the natural isomorphism $\eta : \mathbb{1} \to **$, where η_E is the identity from E to in .

Now that we have constructed a pivotal structure on , the statement of Proposition 2.1 feels intuitive and the proof falls out quite nicely.

Proof (Proposition 2.1). Fix an $E \in$ and let $U = Z(E)$ and $V = Z()$. Let ${}_E : \otimes E \to \emptyset$ and ${}_E : \emptyset \to E\otimes$ be the evaluation and co-evaluation maps mentioned above. We have already shown that $(\mathrm{id}_E \otimes {}_E) \circ ({}_E \otimes \mathrm{id}_E) = \mathrm{id}_E$ in the sense that the following diagram commutes.

$$\begin{tikzcd}
E \arrow[r, "\lambda_E^{-1}"] \arrow[d, "\mathrm{id}_E"] & \emptyset \otimes E \arrow[r, "{}_E \otimes \mathrm{id}_E"] & (E\otimes) \otimes E \arrow[d, "\alpha_{E,,E}"] \\
E & E \otimes \emptyset \arrow[l, "\rho_E"] & E \otimes (\otimes E) \arrow[l, "\mathrm{id}_E \otimes {}_E"]
\end{tikzcd}$$

Here α is the associator and λ, ρ are the natural isomorphisms associated with the tensor unit in . Similarly, we have that $({}_E \otimes \mathrm{id}_{}) \circ (\mathrm{id}_{} \otimes {}_E) = \mathrm{id}$ in the same sense. Define $T = Z({}_E) : V \otimes U \to$ and $S = Z({}_E) : \to U \otimes V$. Since Z is a symmetric monoidal functor, we have that $(\mathrm{id}_U \otimes T) \circ (S \otimes \mathrm{id}_U) = \mathrm{id}_U$ in the sense that the diagram below commutes.

$$\begin{tikzcd}
U \arrow[r, "\lambda_U'^{-1}"] \arrow[d, "\mathrm{id}_U"] & \otimes U \arrow[r, "S \otimes \mathrm{id}_U"] & (U \otimes V) \otimes U \arrow[d, "\alpha'_{U,V,U}"] \\
U & U \otimes \arrow[l, "\rho'_U"] & U \otimes (V \otimes U) \arrow[l, "\mathrm{id}_U \otimes T"]
\end{tikzcd}$$

Where α' is the associator and λ', ρ' are the natural isomorphisms associated with the tensor unit in . Similarly, we have that $(T \otimes \mathrm{id}_V) \circ (\mathrm{id}_V \otimes S) = \mathrm{id}_V$.

Let $\{u_\alpha\}_{\alpha \in A}, \{v_\beta\}_{\beta \in B}$ be a basis for U and V respectively (note that these may be infinite, even uncountably infinite, but every element is still a finite linear combination of elements in these bases). Then $\{u_\alpha \otimes v_\beta\}_{\alpha \in A, \beta \in B}$ is a basis for $U \otimes V$, so let $S(1) = \sum_{i=1}^N u_i \otimes v_i$ for some $N \in \mathbf{N}$ (note that N is finite). Since $(\mathrm{id}_E \otimes {}_E) \circ ({}_E \otimes \mathrm{id}_E) = \mathrm{id}_E$, for any $u \in U$, we have from following the chain of compositions in the above commutative diagram that

$$u = \sum_{i=1}^{N} T(v_i, u) u_i.$$

That is, we have shown that U is spanned by $\{u_i\}_{i=1,\ldots,N}$ and so U is finite dimensional. Since Z is a symmetric monoidal functor, we have that it takes duals in to duals in and so $Z() \cong Z(E)^*$. $\square$

Immediately from Proposition 2.1 we can prove a nice fact about how to calculate the dimension of $Z(E)$.

Corollary 2.2 *For any $E \in$, $Z(E \times S^1) = \dim_{(} Z(E)) \cdot id$.*[1]

Proof. We can write $E \times S^1$ as a combination of ${}_{E}, {}_{E}$ and the braiding in , as shown below.

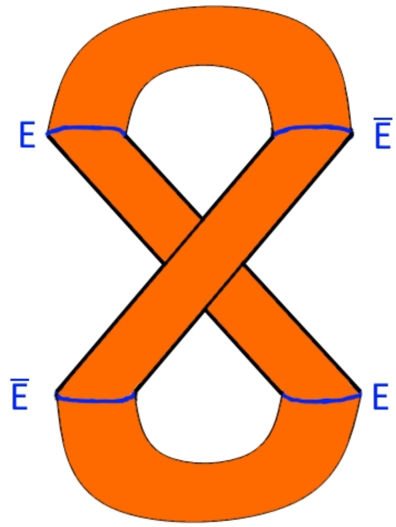

Figure 4: Here we have twisted our $E \times S^1$ into a combination of ev, coev and the braiding.

This composition of maps shows that $E \times S^1 =_E \circ \beta_E \circ_E$ where β_E is the braiding in . Applying Z to this composition we have that

$$Z(E \times S^1) = Z(_E) \circ \beta'_U \circ Z(_E).$$

Where β'_U is the braiding in . So the map $Z(E \times S^1)$ can be calculated by observing where $1 \in$ is mapped to in the following square.

$$\begin{tikzcd}
{} \arrow[r, "Z(_E)"] \arrow[d, dashed, "Z(E\times S^1)"'] & U \otimes U^* \arrow[d, "\beta'_U"] \\
{} & U^* \otimes U \arrow[l, "Z(_E)"]
\end{tikzcd}$$

Since we showed above that $Z(E)$ is finite dimensional and $Z() \cong Z(E)^*$, we have that $Z(_E) :\to U \otimes U^*$ and $Z(_E) : U^* \otimes U \to$ are the co-evaluation and evaluation maps in respectively. So fix a basis $\{u_i\}_{i=1,\ldots,N}$ for U and the corresponding dual basis $\{\phi_i\}_{i=1,\ldots,N}$ for U^* where N is the dimension of U. Then we have that

$$Z(_E)(1) = \sum_{i=1}^{N} u_i \otimes \phi_i$$
$$Z(_E)(\phi_i \otimes u_j) = \delta_{ij}.$$

Thus calculating $Z(E \times S^1)(1)$ gives

$$\begin{aligned} Z(E \times S^1)(1) &= Z(_E)\left(\sum_{i=1}^{N} \phi_i \otimes u_i\right) \\ &= \sum_{i=1}^{N} \delta_{ii} \\ &= N. \end{aligned}$$

We have shown that the map $Z(E \times S^1)$ is multiplication by N and therefore $Z(E \times S^1) = \dim_{(}Z(E)) \cdot \mathrm{id}$. $\square$

By picking nice properties such as being closed and having an orientation on our manifolds, we have been able to show that has a nice pivotal, rigid structure. Further, we have shown that any TQFT must be well behaved in the sense that it only maps to finite dimensional vector spaces and maps duals in to duals in .

3 More Properties

Another good property we are able to show is that if we are given a TQFT $Z :\to$ and an r-manifold X, then we can construct a new $((n-r)+1)$-dimensional TQFT.

Proposition 3.1 *Let $Z :\to$ be a TQFT and X a closed oriented r-manifold with $r < n$. Then we have a TQFT $Z^{:}_{n-r} \to$ defined on objects by $Z^{(}E) = Z(E \times X)$ and defined on morphisms $M : E \to F$ by $Z^{(}M) = Z(M \times X)$.*[1]

Proof. There is not much to prove here. All we need to do is show that $(-) \times X$ is a symmetric monoidal functor from $_{n-r} \to$ because then Z is just a composition of symmetric monoidal functors. But this is clear since taking products commutes with disjoint unions and our defined braiding, in the sense that we can see that $(-) \times X$ will satisfy the appropriate commutative diagrams. □

Furthering our investigation of the structure of TQFTs, we find that the following lemma about general symmetric monoidal functors is quite useful.

Lemma 3.2 *Let $F, G : \mathcal{C} \to \mathcal{D}$ be symmetric monoidal functors between rigid, pivotal categories $\mathcal{C}, \mathcal{D}$. If $\eta : F \to G$ is a symmetric monoidal natural transformation between F and G, then η is a natural isomorphism.*[1]

Proof. For any object $C \in \mathcal{C}$, we have that C has a dual C^* because $\mathcal{C}$ is rigid and we have that $C \cong (C^*)^*$ since $\mathcal{C}$ is pivotal. Using this, we note that $G(C) \cong G(C^*)^*$ and $F(C) \cong F(C^*)^*$ by some isomorphisms $\varepsilon_C : G(C) \to G(C^*)^*, \rho_C : F(C) \to F(C^*)^*$, given by the pivotal structure on $\mathcal{C}$ and $\mathcal{D}$. Then we can show that η_C is an isomorphism by defining an inverse $\eta_C^{-1} = \rho_C^{-1} \circ (\eta_{C^*})^* \circ \varepsilon_C$ where $(\eta_{C^*})^* : G(C^*)^* \to F(C^*)^*$ is the dual map of the component of η on C^*. One can check that η_C^{-1} is a two sided inverse of η_C and therefore η is a natural isomorphism. □

Immediately from this general result about rigid, pivotal categories, we obtain the following information about $(n+1)$-dimensional TQFTs.

Corollary 3.3 *The category of $(n+1)$-dimensional TQFTs (with morphisms symmetric monoidal natural transformations) forms a groupoid.*[1]

Proof. Let Z, Z' be $(n+1)$-dimensional TQFTs and $\eta : Z \to Z'$ a symmetric monoidal natural transformation. We have shown before that , are rigid, pivotal categories, hence by Lemma 3.2, η is an isomorphism. Thus every morphism in the category of $(n+1)$-dimensional TQFTs is invertible and so this category forms a groupoid. □

Finally, we will finish by classifying (0+1)-dimensional TQFTs. This classification shows that (0+1)-dimensional TQFTs are quite boring, but they quickly become complicated as dimension increases.

Theorem 3.4 *There is a 1-1 correspondence between isomorphism classes of $(0+1)$-dimensional TQFTs $Z :_0\to$ and isomorphism classes of finite dimensional vector spaces, given by $Z \mapsto Z()$, where is the positively orientated connected 0-manifold.*[1]

Proof. Let $\mathcal{Z}$ be the set of isomorphism classes of $(0+1)$-TQFTs and $\mathcal{V}$ be the set of isomorphism classes of vector spaces. Further, let

$$F : \mathcal{Z} \to \mathcal{V}$$
$$F(Z) = Z()$$

From Proposition 2.1, $Z()$ is finite dimensional which proves one that F is well defined. To see that F has a two sided inverse, fix a finite dimensional vector space V and note that any object $X \in_0$ can be written as $X \cong^{\sqcup m} \sqcup^{\sqcup n}$. So define a TQFT $Z = F^{-1}(V)$ on objects by

$$Z(^{\sqcup m}\sqcup^{\sqcup n}) = V^{\otimes m} \otimes (V^*)^{\otimes n}.$$

To define Z on morphisms, we notice that the morphisms in $_0$ can be generated (as a symmetric monoidal category) by the following strings.

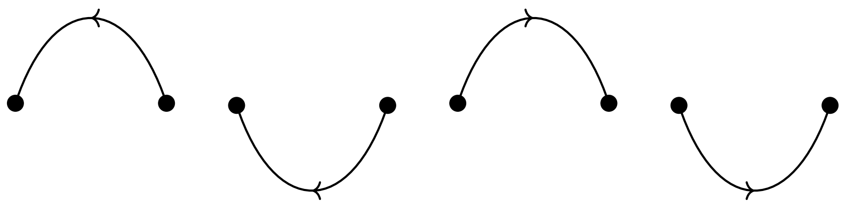

Figure 5: The generating morphisms of $_0$, labelled (from left to right) as M_1, M_2, M_3, M_4.

Proving that these morphisms generate all the morphisms in $_0$ is a long, tedious, and technical procedure and hence has been omitted. These morphisms do look very familiar though and actually correspond to the evaluation and co-evaluation maps in $_0$. Therefore, due to the equivalence relation on the morphisms in $_0$, we also have the following relations on these generators.

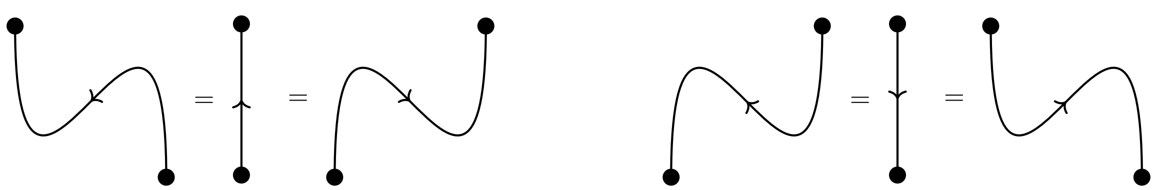

Figure 6: The equivalence relations the morphisms need to satisfy in $_0$.

Again proving that these relations are all the relations of morphisms in $_0$ (including the relations induced by the symmetric braiding) is a long, tedious, and technical procedure and hence has been omitted. Our idea is to map these generating morphisms to the corresponding evaluation and co-evaluation maps in for V. To do this, let $\{e_i\}$ be a basis for V and $\{\phi_i\}$ the corresponding dual basis for V^*. Then define Z on the generators as

$$\begin{aligned} Z(M_1) &: V^* \otimes V \to, & \phi_i \otimes e_j &\mapsto \phi_i(e_j) \\ Z(M_2) &: \to V \otimes V^*, & \lambda &\mapsto \sum_i \lambda e_i \otimes \phi_i \\ Z(M_3) &: V \otimes V^* \to, & e_i \otimes \phi_j &\mapsto \phi_j(e_i) \\ Z(M_4) &: \to V^* \otimes V, & \lambda &\mapsto \sum_i \lambda \phi_i \otimes e_i \end{aligned}$$

Recall here that we have labelled the morphisms $M_1, ..., M_4$ from left to right in Figure 5. From these definitions, we can see that Z acting on morphisms respects the relations in Figure 6. Thus we have a well defined TQFT $Z :_0\to$ and this gives us our inverse F^{-1}.

To prove that these are inverses, firstly we have

$$\begin{aligned}(F \circ F^{-1})(V) &= F^{-1}(V)() \\ &= V.\end{aligned}$$

Now consider $(F^{-1} \circ F)(Z) = F^{-1}(Z())$. Let $W = Z()$. We need to show that $F^{-1}(W) \cong Z$. That is, we need to define a natural isomorphism $\eta : F^{-1}(W) \to Z$. Any object $X \in$ can be written as $X \cong^{\sqcup m} \sqcup^{\sqcup n}$ by an isomorphisms $\psi_X : X \to^{\sqcup m} \sqcup^{\sqcup n}$ so we have

$$\begin{aligned}F^{-1}(W)(X) &\cong F^{-1}(W)(^{\sqcup m}\sqcup^{\sqcup n}) \\ &= W^{\otimes m} \otimes (W^*)^{\otimes n} \\ Z(X) &\cong Z(^{\sqcup m}\sqcup^{\sqcup n}) \\ &= W^{\otimes m} \otimes (W^*)^{\otimes n}.\end{aligned}$$

That is, define the components of η as $\eta_X = Z(\psi_X)^{-1} \circ F^{-1}(W)(\psi_X)$. The naturality of η comes from our definition on morphisms that sent evaluation and coevaluation maps to their respective evaluation and coevaluation maps. Thus we have shown that F^{-1} is a two sided inverse of F and this completes our correspondence. $\square$

This classification of $(0 + 1)$-dimensional TQFTs shows that they are boring and can be treated as essentially natural numbers. However, it does leave us hopeful about classifying higher dimensional TQFTs. Even though they become more complicated very quickly as dimension increases, the $(0+1)$-dimensional classification, and the previous groupoid and other properties we proved may hint at some deeper structure about higher dimensional TQFTs and they could potentially be classified with some higher algebraic structures.

References

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*The statements/proof ideas of this paper are from this reference, but what I have written is my own work.